

SOLIDWORKS API (.NET VBA C#) — Reference (199/337): IModelDocExtension (part 5 of 8)

SOLIDWORKS API Documentation (converted from CHM)

IModelDocExtension (part 5 of 8) (Type)

SOLIDWORKS API Help

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HasMaterialPropertyValues Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : HasMaterialPropertyValues Method (IModelDocExtension)

Gets whether this model has an appearance.

.NET Syntax

Visual Basic (Declaration)

```
Function HasMaterialPropertyValues() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean
```

```
value = instance.HasMaterialPropertyValues()
```

C#

```
System.bool HasMaterialPropertyValues()
```

C++/CLI

```
System.bool HasMaterialPropertyValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if this model has an appearance, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::HasMaterialPropertyValues](#).

Example

[Add and Delete Appearances from Specific Display States \(VBA\)](#)
[Add and Delete Appearances from Specific Display States \(VB.NET\)](#)
[Add and Delete Appearances from Specific Display States \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SetMaterialPropertyValues Method \(\)](#)
[IModelDocExtension::RemoveMaterialProperty Method \(\)](#)
[IModelDocExtension::GetMaterialPropertyValues Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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HasRenamedDocuments Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : HasRenamedDocuments Method (IModelDocExtension)

Gets whether the document has renamed files.

.NET Syntax

Visual Basic (Declaration)

```
Function HasRenamedDocuments() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.Boolean

value = instance.HasRenamedDocuments()
```

C#

```
System.bool HasRenamedDocuments()
```

C++/CLI

```
System.bool HasRenamedDocuments();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the document has renamed files, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::HasRenamedDocuments](#).

Example

[Rename Components and Save Assembly \(C#\)](#)
[Rename Components and Save Assembly \(VB.NET\)](#)
[Rename Components and Save Assembly \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::RenameDocument Method \(\)](#)
[IRenamedDocumentReferences Interface](#)

Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

HideDecal Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : HideDecal Method (IModelDocExtension)

Hides or shows the specified decal applied to this model.

.NET Syntax

Visual Basic (Declaration)

```
Function HideDecal( _
    ByVal DecalID As System.Integer, _
    ByVal Hide As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim DecalID As System.Integer
Dim Hide As System.Boolean
Dim value As System.Boolean

value = instance.HideDecal(DecalID, Hide)
```

C#

```
System.bool HideDecal(
    System.int DecalID,
    System.bool Hide
)
```

C++/CLI

```
System.bool HideDecal(
    & System.int DecalID,
    & System.bool Hide
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DecalID

[Decal ID](#)

Hide

True to hide the decal, false to show it

Return Value

True if the decal is hidden or shown, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::HideDecal](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IDecal Interface](#)

[IModelDocExtension::AddDecal Method](#)

[IModelDocExtension::CreateDecal Method](#)

[IModelDocExtension::DeleteAllDecals Method](#)

[IModelDocExtension::DeleteDecal Method](#)

[IModelDocExtension::GetDecal Method](#)

[IModelDocExtension::GetDecals Method](#)

[IModelDocExtension::GetDecalsCount Method](#)

[IModelDocExtension::IGetDecals Method](#)

[IModelDocExtension::MoveDecal Method](#)

[IModelDocExtension::ReverseDecalsOrder Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

HideFeatureManager Method (IModelDocExtension)

See Also [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : HideFeatureManager Method (IModelDocExtension)

Hides or shows the Manager Pane.

.NET Syntax

Visual Basic (Declaration)

```
Function HideFeatureManager( _
    ByVal Hide As System.Boolean _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Hide As System.Boolean  
Dim value As System.Boolean  
  
value = instance.HideFeatureManager(Hide)
```

C#

```
System.bool HideFeatureManager(  
    System.bool Hide  
)
```

C++/CLI

```
System.bool HideFeatureManager(  
    & System.bool Hide  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Hide

True to hide the Manager Pane, false to not

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::HideFeatureManager](#).

Example

[Hide the FeatureManager Design Tree \(VBA\)](#)
[Hide the FeatureManager Design Tree \(VB.NET\)](#)
[Hide the FeatureManager Design Tree \(C#\)](#)
[Change Active Tab in Manager Pane \(C#\)](#)
[Change Active Tab in Manager Pane \(VB.NET\)](#)
[Change Active Tab in Manager Pane \(VBA\)](#)

Remarks

The FeatureManager design tree, PropertyManager, ConfigurationManager, DimXpertManager, DisplayManager, and custom tabs reside in the Manager Pane.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::FeatureManagerFilterString Property](#)
[IModelViewManager::FeatureManagerTabsVisible Property\(\)](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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IAddDisplayStateSpecificRenderMaterial Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IAddDisplayStateSpecificRenderMaterial Method (IModelDocExtension)

Adds the specified material to the specified display states in the active configuration and returns the IDs assigned to that material.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddDisplayStateSpecificRenderMaterial( _
    ByVal PRenderMaterial As RenderMaterial, _
    ByVal DisplayStateOption As System.Integer, _
    ByVal DisplayStateCount As System.Integer, _
    ByRef DisplayStateNames As System.String, _
    ByRef PWMaterialId1 As System.Integer, _
    ByRef PWMaterialId2 As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim PRenderMaterial As RenderMaterial
Dim DisplayStateOption As System.Integer
Dim DisplayStateCount As System.Integer
Dim DisplayStateNames As System.String
Dim PWMaterialId1 As System.Integer
Dim PWMaterialId2 As System.Integer
Dim value As System.Boolean
```

```
value = instance.IAddDisplayStateSpecificRenderMaterial(PRenderMaterial, DisplayStateOption,
DisplayStateCount, DisplayStateNames, PWMaterialId1, PWMaterialId2)
```

C#

```
System.bool IAddDisplayStateSpecificRenderMaterial(
    RenderMaterial PRenderMaterial,
    System.int DisplayStateOption,
    System.int DisplayStateCount,
    ref System.string DisplayStateNames,
    out System.int PWMaterialId1,
    out System.int PWMaterialId2
)
```

C++/CLI

```
System.bool IAddDisplayStateSpecificRenderMaterial(
    & RenderMaterial^ PRenderMaterial,
    & System.int DisplayStateOption,
    & System.int DisplayStateCount,
    & System.String^% DisplayStateNames,
    & [Out] System.int PWMaterialId1,
    & [Out] System.int PWMaterialId2
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PRenderMaterial

[Material](#) to add

DisplayStateOption

Display states as defined in [swDisplayStateOptions](#)

DisplayStateCount

Number of display states (see **Remarks**)

DisplayStateNames

- in-process, unmanaged C++: Pointer to an array of the names of the display states to which to add material
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

PWMaterialId1

First ID of material

PWMaterialId2

Second ID of material

Return Value

True if material is added, false if not

Remarks

Before calling this method, call [IConfiguration::GetDisplayStatesCount](#) to get DisplayStateCount.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::IDeleteDisplayStateSpecificRenderMaterial Method](#)
[IModelDocExtension::AddDisplayStateSpecificRenderMaterial Method](#)
[IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial Method](#)
[IRenderMaterial Interface](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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IChangeSketchPlane Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IChangeSketchPlane Method (IModelDocExtension)

Changes the plane used by a sketch by moving the selected sketch to the selected plane in the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function IChangeSketchPlane( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Boolean  
  
value = instance.IChangeSketchPlane(Config_opt, Config_count, Config_names)
```

C#

```
System.bool IChangeSketchPlane(  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.bool IChangeSketchPlane(  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configurations as defined by [swInConfigurationOpts_e](#)
Config_count
Number of configurations
Config_names
Array of configuration names

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IChangeSketchPlane](#).

See Also

ICreateOLEObject Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ICreateOLEObject Method (IModelDocExtension)

Creates an OLE object on the active document.

.NET Syntax

Visual Basic (Declaration)

```
Function ICreateOLEObject( _  
    ByVal Aspect As System.Integer, _  
    ByRef Position As System.Double, _  
    ByVal ByteCount As System.Integer, _  
    ByRef Buffer As System.Byte, _  
    ByRef ErrorCode As System.Integer _  
) As SwOLEObject
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Aspect As System.Integer  
Dim Position As System.Double  
Dim ByteCount As System.Integer  
Dim Buffer As System.Byte  
Dim ErrorCode As System.Integer  
Dim value As SwOLEObject  
  
value = instance.ICreateOLEObject(Aspect, Position, ByteCount, Buffer, ErrorCode)
```

C#

```
SwOLEObject ICreateOLEObject(  
    System.int Aspect,  
    ref System.double Position,  
    System.int ByteCount,  
    ref System.byte Buffer,  
    out System.int ErrorCode  
)
```

C++/CLI

```
SwOLEObject^ ICreateOLEObject(  
& System.int Aspect,  
& System.double% Position,  
& System.int ByteCount,  
& System.byte% Buffer,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Aspect

Viewing aspect of the OLE object as defined in the DVASPECT enumeration (see **Remarks**)

Position

Top-left and bottom-right positions (see **Remarks**)

ByteCount

Number of bytes

Buffer

Data for the OLE object (see **Remarks**)

ErrorCode

0 if True or 1 if false

Return Value

[OLE object](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ICreateOLEObject](#).

Remarks

Argument	Information
Aspect	Uses the DVASPECT enumeration, which has the following values: - DVASPECT_CONTENT = 1 - DVASPECT_THUMBNAIL = 2 - DVASPECT_ICON = 4 - DVASPECT_DOCPRINT = 8 See the MSDN documentation for details about the DVASPECT enumeration.
Position	Specify: - Sheet coordinates for drawings. - Screen pixel coordinates for parts and assemblies.
Buffer	See ISwOLEObject::Buffer or specify ISwOLEObject::IGetBuffer - or - Get the data from your own OLE object. The data is in the format obtained from the Microsoft MFC object COleClientItem using the GetHGlobalFromILockBytes.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::CreateOLEObject Method](#)
[IModelDocExtension::GetOLEObjectCount Method](#)
[IModelDocExtension::GetOLEObjects Method](#)
[IModelDocExtension::IGetOLEObjects Method](#)
[IModelDocExtension::InsertObjectFromFile Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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IDeleteDisplayStateSpecificRenderMaterial Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IDeleteDisplayStateSpecificRenderMaterial Method (IModelDocExtension)

Deletes the specified materials, using the IDs of the materials, from the active configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function IDeleteDisplayStateSpecificRenderMaterial( _  
    ByVal IdCount As System.Integer, _  
    ByRef PWMaterialId1 As System.Integer, _  
    ByRef PWMaterialId2 As System.Integer _  
    ) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim IdCount As System.Integer  
Dim PWMaterialId1 As System.Integer  
Dim PWMaterialId2 As System.Integer  
Dim value As System.Boolean
```



```
value = instance.IDeleteDisplayStateSpecificRenderMaterial(IdCount, PWMaterialId1, PWMaterialId2)
```

C#

```
System.bool IDeleteDisplayStateSpecificRenderMaterial(  
    System.int IdCount,  
    ref System.int PWMaterialId1,  
    ref System.int PWMaterialId2  
)
```

C++/CLI

```
System.bool IDeleteDisplayStateSpecificRenderMaterial(  
& System.int IdCount,  
& System.int% PWMaterialId1,  
& System.int% PWMaterialId2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IdCount

Number of material IDs

PWMaterialId1

- in-process, unmanaged C++: Pointer to an array of the first IDs of the material to delete

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

PWMaterialId2

- in-process, unmanaged C++: Pointer to an array of the second IDs of the material to delete

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the materials are deleted, false if not

Remarks

Before calling this method, call:

- [IModelDocExtension::GetRenderMaterialsCount](#) to get the value for IdCount.
- [IRenderMaterial::GetMaterialIds](#) to get the values for PWMaterialId1 and PWMaterialId2, the IDs of the materials added to the model.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::AddDisplayStateSpecificRenderMaterial Method](#)

[IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial Method](#)

[IModelDocExtension::AddDisplayStateSpecificRenderMaterial Method](#)

[IRenderMaterial Interface](#)

Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

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IEditDimensionProperties Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IEditDimensionProperties Method (IModelDocExtension)

Edits the selected dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function IEditDimensionProperties( _  
    ByVal TolType As System.Integer, _  
    ByVal TolMax As System.Double, _  
    ByVal TolMin As System.Double, _  
    ByVal TolMaxFit As System.String, _
```

```

ByVal TolMinFit As System.String, _
ByVal UseDocPrec As System.Boolean, _
ByVal Precision As System.Integer, _
ByVal ArrowsIn As System.Integer, _
ByVal UseDocArrows As System.Boolean, _
ByVal Arrow1 As System.Integer, _
ByVal Arrow2 As System.Integer, _
ByVal PrefixText As System.String, _
ByVal SuffixText As System.String, _
ByVal ShowValue As System.Boolean, _
ByVal CalloutText1 As System.String, _
ByVal CalloutText2 As System.String, _
ByVal DimensionLowerText As System.String, _
ByVal CenterText As System.Boolean, _
ByVal ConfigOption As System.Integer, _
ByVal Count As System.Integer, _
ByRef ConfigNames As System.String _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim TolType As System.Integer
Dim TolMax As System.Double
Dim TolMin As System.Double
Dim TolMaxFit As System.String
Dim TolMinFit As System.String
Dim UseDocPrec As System.Boolean
Dim Precision As System.Integer
Dim ArrowsIn As System.Integer
Dim UseDocArrows As System.Boolean
Dim Arrow1 As System.Integer
Dim Arrow2 As System.Integer
Dim PrefixText As System.String
Dim SuffixText As System.String
Dim ShowValue As System.Boolean
Dim CalloutText1 As System.String
Dim CalloutText2 As System.String
Dim DimensionLowerText As System.String
Dim CenterText As System.Boolean
Dim ConfigOption As System.Integer
Dim Count As System.Integer
Dim ConfigNames As System.String
Dim value As System.Boolean

```

```

value = instance.IEditDimensionProperties(TolType, TolMax, TolMin, TolMaxFit, TolMinFit,
UseDocPrec, Precision, ArrowsIn, UseDocArrows, Arrow1, Arrow2, PrefixText, SuffixText, ShowValue,
CalloutText1, CalloutText2, DimensionLowerText, CenterText, ConfigOption, Count, ConfigNames)

```

C#

```

System.bool IEditDimensionProperties(
    System.int TolType,
    System.double TolMax,
    System.double TolMin,
    System.string TolMaxFit,
    System.string TolMinFit,
    System.bool UseDocPrec,
    System.int Precision,
    System.int ArrowsIn,
    System.bool UseDocArrows,
    System.int Arrow1,
    System.int Arrow2,
    System.string PrefixText,
    System.string SuffixText,
    System.bool ShowValue,
    System.string CalloutText1,
    System.string CalloutText2,
    System.string DimensionLowerText,

```

```

        System.bool CenterText,
        System.int ConfigOption,
        System.int Count,
        ref System.string ConfigNames
    )

```

C++/CLI

```

System.bool IEditDimensionProperties(
    & System.int TolType,
    & System.double TolMax,
    & System.double TolMin,
    & System.String^ TolMaxFit,
    & System.String^ TolMinFit,
    & System.bool UseDocPrec,
    & System.int Precision,
    & System.int ArrowsIn,
    & System.bool UseDocArrows,
    & System.int Arrow1,
    & System.int Arrow2,
    & System.String^ PrefixText,
    & System.String^ SuffixText,
    & System.bool ShowValue,
    & System.String^ CalloutText1,
    & System.String^ CalloutText2,
    & System.String^ DimensionLowerText,
    & System.bool CenterText,
    & System.int ConfigOption,
    & System.int Count,
    & System.String^% ConfigNames
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TolType

Type of tolerance as defined in [swTolType_e](#)

TolMax

Maximum value for the tolerance

TolMin

Minimum value for the tolerance

TolMaxFit

Text for the maximum FIT value when using a fit tolerance type

TolMinFit

Text for the minimum FIT value when using a fit tolerance type

UseDocPrec

True to use the document's precision value, false to use value specified for Precision

Precision

Precision to use for this dimension if UseDocPrec is false

ArrowsIn

Arrow direction as defined in [swDimensionArrowsSide_e](#)

UseDocArrows

True to use the document's arrow types, false to use values specified for Arrow1 and Arrow2

Arrow1

Type of arrow to use for the first arrow of this dimension as defined in [swArrowStyle_e](#) if UseDocArrows is false

Arrow2

Type of arrow to use for the second arrow of this dimension as defined in [swArrowStyle_e](#) if UseDocArrows is false

PrefixText

Text for the prefix of the dimension

SuffixText

Text for the suffix of the dimension

ShowValue

True to display the value of the dimension in the user interface, false to not

CalloutText1

Callout text to display above the dimension

CalloutText2

Callout text to display below the dimension

DimensionLowerText

Text to display below the dimension line; valid for [display_dimensions](#) in drawings only

CenterText

True to center the text in the dimension, false to not

ConfigOption

Configuration option as defined in [swInConfigurationOpts_e](#)

Count

Number of configurations

ConfigNames

- in-process, unmanaged C++: Pointer to an array of configuration names to which to apply these edits (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

True if the dimension is modified, false if not

Remarks

This method:

- always changes the dimension's parameters in the active configuration. For example, if you specify `swInConfigurationOpts_e.swSpecifyConfiguration` for `ConfigOption` and do not specify the name of the active configuration in `ConfigNames`, then the dimension parameters in the active configuration are affected.
- does not support [hole callouts](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::EditDimensionProperties Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGet3rdPartyStorageStore Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGet3rdPartyStorageStore Method (IModelDocExtension)

Gets the IStorage interface to the specified third-party storage in this SOLIDWORKS document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGet3rdPartyStorageStore( _  
    ByVal SubStorageName As System.String, _  
    ByVal IsStoring As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim SubStorageName As System.String  
Dim IsStoring As System.Boolean  
Dim value As System.Object  
  
value = instance.IGet3rdPartyStorageStore(SubStorageName, IsStoring)
```

C#

```
System.object IGet3rdPartyStorageStore(  
    System.string SubStorageName,  
    System.bool IsStoring  
)
```

C++/CLI

```
System.Object^ IGet3rdPartyStorageStore(  
& System.String^ SubStorageName,  
& System.bool IsStoring  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SubStorageName
Name of the storage
IsStoring
True if you are storing data, false if not

Return Value

Unknown type, the third-party IStorage storage

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGet3rdPartyStorageStore](#).

Remarks

Call this interface when a `SaveToStorageStoreNotify` or `LoadFromStorageStoreNotify` notification is received. Do not call this method to save or load from storage in reaction to the `FileSaveNotify` or `FileOpenNotify` events.

During a file-open operation, your application receives a `LoadFromStorageStoreNotify` event if you have registered for document notifications. When you receive this event, it is safe to open your `Stream` for reading using `ModelDocExtension::IGet3rdPartyStorageStore`. When a file is fully loaded, you can call this method for reading your data at any time. This provides you with access to read your storage node whenever needed. However, it is not a good idea to attempt this while in the middle of a `FileSaveNotify` or `FileOpenNotify` as unforeseen conflicts can occur.

If your application is loaded in the middle of a SOLIDWORKS Design session, then you can traverse all open documents using [ISldworks::EnumDocuments2](#), [ISldworks::IGetFirstDocument2](#), or [IModelDoc2::IGetNext](#), and read the storage from all existing open documents. From that point on, your application can use the `LoadFromStorageStoreNotify` event to recognize when new documents have been opened and need reading.

For storage writing, `IModelDoc2::IGet3rdPartyStorageStore` interface is locked unless the `SaveToStorageStoreNotify` event has been sent. Therefore, you can only write to your storage when the file is actually being saved and your application has received the `SaveToStorageStoreNotify` event.

Your call to `IModelDocExtension::IGet3rdPartyStorageStore` must be followed by a call to [IModelDocExtension::IRelease3rdPartyStorageStore](#). This must be done even when you fail to obtain a `Stream` and the return value from `IModelDocExtension::IGet3rdPartyStorageStore` is `NULL`. If you fail to call `IModelDocExtension::IRelease3rdPartyStorageStore`, the third-party node may remain locked and prevent future access. You are not required to call `IModelDocExtension::IRelease3rdPartyStorageStore`, under any circumstance, if you called `IModelDocExtension::IGet3rdPartyStorageStore` in reaction to one of the `SaveToStorageStoreNotify` or `LoadFromStorageStoreNotify` events. However, calling `IModelDocExtension::IRelease3rdPartyStorageStore` excessively does not cause problems.

As work progresses during an active session and you have information that needs to be stored, you may want to flag the document as dirty using [IModelDoc2::SetSaveFlag](#). This causes a "Save Changes?" prompt to display if the user tries to close the file without saving. If the user clicks **Yes** to this prompt, then your application receives the `SaveToStorageStoreNotify` event.

NOTE: The name given to the storage should be registered with so that no conflicts occur. Once registered, the storage name is reserved exclusively for your application.

Passing the unique ID string and a flag to determine if data is being stored or loaded returns an `IUnknown` pointer. You must then use `QueryInterface()` to get the **IStorage** interface. The storage is used for serialization and then released in the third-party code.

The `IStorage` object used by the third party is written to `IStorage` objects under an `IStorage` object called `ThirdPtyStore` in the compound document. Each third party writes to a single `IStorage` whose name is assigned by SOLIDWORKS.

```
SwRootStorage --|
|
|-- ThirdPtyStore --|
|
|
|-- <SW Assigned IStorage name 1> --|
|
|           |-- <Application 1 IStream name 1>
|           |-- <Application 1 IStream name 2>
|
|
|-- <SW Assigned IStorage name 2> --|
|
|           |-- <Application 2 IStream name 1>
|           |-- <Application 2 IStream name 2>
|           |-- <Application 2 IStorage name 2> --|
|
|           |-- <Application 2 IStream name 3>
```

NOTE: If you are using serialization, then be careful with the standard MFC macros; otherwise, you may get a message like "Unexpected File Format" after your application is unloaded. One way of using `IMPLEMENT_SERIAL` is:

```
IMPLEMENT_SERIAL( CCustomAttr, CObject, VERSIONABLE_SCHEMA|0 )
```

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

IGetAnnotations Method (IModelDocExtension)

[See Also](#)



.NET Syntax

Visual Basic (Declaration)

```
Function IGetAnnotations( _  
    ByVal NumAnnotations As System.Integer _  
) As Annotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim NumAnnotations As System.Integer  
Dim value As Annotation  
  
value = instance.IGetAnnotations(NumAnnotations)
```

C#

```
Annotation IGetAnnotations(  
    System.int NumAnnotations  
)
```

C++/CLI

```
Annotation^ IGetAnnotations(  
&    System.int NumAnnotations  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumAnnotations
Number of annotations

Return Value

- in-process, unmanaged C++: Pointer to an array of [annotations](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [IModelDocExtension::GetAnnotationCount](#) before calling this method.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetAnnotations Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAnnotationViews Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetAnnotationViews Method (IModelDocExtension)

Gets the annotation views in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAnnotationViews( _  
    ByVal AnnotationViewCount As System.Integer _  
) As AnnotationView
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim AnnotationViewCount As System.Integer  
Dim value As AnnotationView  
  
value = instance.IGetAnnotationViews(AnnotationViewCount)
```

C#

```
AnnotationView IGetAnnotationViews(  
    System.int AnnotationViewCount  
)
```

C++/CLI

```
AnnotationView^ IGetAnnotationViews(  
& System.int AnnotationViewCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AnnotationViewCount

Number of annotation views in this part or assembly document

Return Value

- in-process, unmanaged C++: Pointer to an array of [annotation views](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IModelDocExtension::AnnotationViewCount](#) to get the value of AnnotationViewCount.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::AnnotationViews Property](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetAttachments Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetAttachments Method (IModelDocExtension)

Gets the attachments for this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetAttachments( _  
    ByVal NumAttachments As System.Integer, _  
    ByRef LinkedArr As System.Boolean _  
) As System.String
```

Visual Basic (Usage)




```

Dim instance As IModelDocExtension
Dim NumAttachments As System.Integer
Dim LinkedArr As System.Boolean
Dim value As System.String

value = instance.IGetAttachments(NumAttachments, LinkedArr)

```

C#

```

System.string IGetAttachments(
    System.int NumAttachments,
    out System.bool LinkedArr
)

```

C++/CLI

```

System.String^ IGetAttachments(
    & System.int NumAttachments,
    & [Out] System.bool LinkedArr
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumAttachments

Number of attachments for this document

LinkedArr

- in-process, unmanaged C++: Pointer to an array of VARIANT_BOOL values indicating if a document is linked or not

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

- in-process, unmanaged C++: Pointer to an array of strings of the names of the attachments for this document

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

This method is supported by SOLIDWORKS 2005 and later.

Call [IModelDocExtension::GetAttachmentCount](#) before calling this method to determine the size of the array.

There is a one-to-one correspondence between the output arrays.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::GetAttachments Method](#)

[IModelDocExtension::DeleteAttachment Method](#)

[IModelDocExtension::InsertAttachment Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetDecals Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetDecals Method (IModelDocExtension)

Gets the decals applied to the model.

.NET Syntax

Visual Basic (Declaration)

```

Function IGetDecals( _
    ByVal Count As System.Integer _
) As Decal

```

```
Dim instance As IModelDocExtension
Dim Count As System.Integer
Dim value As Decal

value = instance.IGetDecals(Count)
```

C#

```
Decal IGetDecals(
    System.int Count
)
```

C++/CLI

```
Decal^ IGetDecals(
    & System.int Count
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of decals

Return Value

- in-process, unmanaged C++: Pointer to an array of [decals](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IModelDocExtension::GetDecalsCount](#) to get the value of Count.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDecal Interface](#)
[IModelDocExtension::AddDecal Method](#)
[IModelDocExtension::CreateDecal Method](#)
[IModelDocExtension::DeleteAllDecals Method](#)
[IModelDocExtension::DeleteDecal Method](#)
[IModelDocExtension::GetDecal Method](#)
[IModelDocExtension::GetDecals Method](#)
[IModelDocExtension::HideDecal Method](#)
[IModelDocExtension::MoveDecal Method](#)
[IModelDocExtension::ReverseDecalsOrder Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetFlattenSheetMetalPersistReference Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetFlattenSheetMetalPersistReference Method (IModelDocExtension)

Gets a byte array of persistent reference IDs for the specified entity (a face, edge, or vertex) in a flattened sheet metal part.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetFlattenSheetMetalPersistReference( _
    ByVal DispObj As System.Object, _
    ByVal Count As System.Integer _
) As System.Byte
```

Visual Basic (Usage)

```

Dim instance As IModelDocExtension
Dim DispObj As System.Object
Dim Count As System.Integer
Dim value As System.Byte

value = instance.IGetFlattenSheetMetalPersistReference(DispObj, Count)

```

C#

```

System.byte IGetFlattenSheetMetalPersistReference(
    System.object DispObj,
    System.int Count
)

```

C++/CLI

```

System.byte IGetFlattenSheetMetalPersistReference(
    & System.Object^ DispObj,
    & System.int Count
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DispObj
Entity (a face, edge, or vertex) in the flattened sheet metal part whose persistent reference IDs you want

Count
Number of persistent reference IDs

Return Value

- in-process, unmanaged C++: Pointer to a byte array of persistent reference IDs
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetFlattenSheetMetalPersistReference](#).

Remarks

The sheet metal part must be in the flattened state when this method is called. You can pass the byte array of persistent reference IDs to [IModelDocExtension::GetSheetMetalObjectsByPersistReference](#) or [IModelDocExtension::IGetSheetMetalObjectsByPersistReference](#), which gets the objects that correspond to the persistent reference IDs in the folded configuration of the sheet metal part.

See [Persistent Reference IDs](#) for more information.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetFlattenSheetMetalPersistReference Method](#)
[IModelDocExtension::GetObjectByPersistReference3 Method](#)
[IModelDocExtension::GetPersistReference3 Method](#)
[IModelDocExtension::GetPersistReferenceCount Method](#)
[IModelDocExtension::IGetObjectByPersistReference3 Method](#)
[IModelDocExtension::IGetPersistReference3 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMassProperties Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetMassProperties Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::GetMassProperties2](#).

Visual Basic (Declaration)

```
Function IGetMassProperties( _
    ByVal Accuracy As System.Integer, _
    ByRef Status As System.Integer _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim Accuracy As System.Integer
Dim Status As System.Integer
Dim value As System.Double

value = instance.IGetMassProperties(Accuracy, Status)
```

C#

```
System.double IGetMassProperties(
    System.int Accuracy,
    out System.int Status
)
```

C++/CLI

```
System.double IGetMassProperties(
    & System.int Accuracy,
    & [Out] System.int Status
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Accuracy

- 0 = as is
- 1 = default
- 2 = maximum

Status

Status of the mass property results as defined in [swMassPropertiesStatus_e](#)

Return Value

- in-process, unmanaged C++: Pointer to an array of size 13; the last element is the accuracy at which returned mass properties are calculated (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

The return value is a 0-based array of doubles as follows:

[*CenterOfMassX*, *CenterOfMassY*, *CenterOfMassZ*, *Volume*, *Area*, *Mass*, *MomXX*, *MomYY*, *MomZZ*, *MomXY*, *MomZX*, *MomYZ*, *Accuracy*]

You can use [ISldWorks::GetUserPreferenceDoubleValue](#) and [swMaterialPropertyDensity](#) to get the density of the SOLIDWORKS part. If the user did not explicitly set the density, then SOLIDWORKS uses a value of 1.0. You can also derive the density of the body by calculating:

$$\text{Density} = (\text{Mass} / \text{Volume})$$

This method returns metric units unless explicitly specified otherwise.

If this object is...

Then...

An assembly

- SOLIDWORKS does not include any suppressed components in the mass property analysis. See [IComponent2::GetSuppression](#) to determine the state of each assembly component.
- This method returns the moments of inertia (MOI) about the assembly center-of-gravity coordinate system aligned with the assembly axes.

A part

The calculated origin for the returned values are based on the default coordinate systems of the model document. They are not based on the selected coordinate system.

This method supports multibody parts.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::CreateMassProperty Method](#)
[IModelDocExtension::GetMassProperties Method](#)
[IMassProperty Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetMaterialPropertyValues Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetMaterialPropertyValues Method (IModelDocExtension)

Gets the material properties for this model.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetMaterialPropertyValues( _  
    ByVal Config\_opt As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Config_opt As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Double  
  
value = instance.IGetMaterialPropertyValues(Config_opt, Config_count, Config_names)
```

C#

```
System.double IGetMaterialPropertyValues(  
    System.int Config\_opt,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.double IGetMaterialPropertyValues(  
& System.int Config\_opt,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_opt
Configuration options as defined by [swInConfigurationOpts_e](#)

Config_count
Number of configurations for this component

Config_names

- In-process, unmanaged C++: Pointer to an array of configuration names for this component
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

- in-process, unmanaged C++: Points to an array of material values for this component (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

The material values returned include the face color (R,G,B values), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

The format of material_values is an array of doubles:

[*R, G, B, Ambient, Diffuse, Specular, Shininess, Transparency, Emission*]

Valid values are from 0 to 1 for all variables. If material_values is all -1 values, then material property values were not assigned to the feature. Therefore, the feature will have the default properties in the user interface.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::IRemoveMaterialProperty Method](#)
[IModelDocExtension::ISetMaterialPropertyValues Method](#)
[IModelDocExtension::RemoveMaterialProperty Method](#)
[IModelDocExtension::SetMaterialPropertyValues Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetModelViews Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetModelViews Method (IModelDocExtension)

Gets the model views in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetModelViews( _  
    ByVal NumModelViews As System.Integer _  
) As ModelView
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim NumModelViews As System.Integer  
Dim value As ModelView  
  
value = instance.IGetModelViews(NumModelViews)
```

C#

```
ModelView IGetModelViews(  
    System.int NumModelViews  
)
```

C++/CLI

```
ModelView^ IGetModelViews(  
& System.int NumModelViews  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumModelViews
Number of model views in this document

Return Value

- in-process, unmanaged C++: Pointer to an array of [model views](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IModelDocExtension::GetModelViewCount](#) to get NumModelViews.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetModelViews Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelView::GetNext Method](#)
[IModelView::IGetNext Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetNamedViewRotation Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetNamedViewRotation Method (IModelDocExtension)

Gets the specified named view orientation matrix with respect to the Front view.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function IGetNamedViewRotation( _  
    ByVal Name As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Name As System.String  
Dim value As System.Double  
  
value = instance.IGetNamedViewRotation(Name)
```

C#

```
System.double IGetNamedViewRotation(  
    System.string Name  
)
```

C++/CLI

```
System.double IGetNamedViewRotation(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Name of the named view

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetNamedViewRotation](#).

Remarks

Array of 9 doubles describing the view rotation with respect to the Front view

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::IModelDocExtension::GetNamedViewRotation Method](#)
[IModelDoc2::DeleteNamedView Method](#)
[IModelDoc2::GetStandardViewRotation Method](#)
[IModelDoc2::IGetStandardViewRotation Method](#)
[IModelDoc2::NameView Method](#)
[IModelDoc2::ShowNamedView2 Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetObjectByPersistReference Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetObjectByPersistReference Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::IGetObjectByPersistReference3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetObjectByPersistReference( _  
    ByVal Count As System.Integer, _  
    ByRef PersistId As System.Byte _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Count As System.Integer  
Dim PersistId As System.Byte  
Dim value As System.Object  
  
value = instance.IGetObjectByPersistReference(Count, PersistId)
```

C#

```
System.object IGetObjectByPersistReference(  
    System.int Count,  
    ref System.byte PersistId  
)
```

C++/CLI

```
System.Object^ IGetObjectByPersistReference(  
& System.int Count,  
& System.byte% PersistId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

PersistId

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetObjectByPersistReference](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetObjectByPersistReference2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetObjectByPersistReference2 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::IGetObjectByPersistReference3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetObjectByPersistReference2( _  
    ByVal Count As System.Integer, _  
    ByRef PersistId As System.Byte, _  
    ByRef ErrorCode As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Count As System.Integer  
Dim PersistId As System.Byte  
Dim ErrorCode As System.Integer  
Dim value As System.Object  
  
value = instance.IGetObjectByPersistReference2(Count, PersistId, ErrorCode)
```

C#

```
System.object IGetObjectByPersistReference2(  
    System.int Count,  
    ref System.byte PersistId,  
    out System.int ErrorCode  
)
```

C++/CLI

```
System.Object^ IGetObjectByPersistReference2(  
& System.int Count,  
& System.byte% PersistId,  
& [Out] System.int ErrorCode  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

PersistId

ErrorCode

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetObjectByPersistReference2](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

IGetObjectByPersistReference3 Method (IModelDocExtension)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetObjectByPersistReference3 Method (IModelDocExtension)

Gets the object assigned to the specified persistent reference ID.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetObjectByPersistReference3( _
    ByVal Count As System.Integer, _
    ByRef PersistId As System.Byte, _
    ByRef ErrorCode As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim Count As System.Integer
Dim PersistId As System.Byte
Dim ErrorCode As System.Integer
Dim value As System.Object

value = instance.IGetObjectByPersistReference3(Count, PersistId, ErrorCode)
```

C#

```
System.object IGetObjectByPersistReference3(
    System.int Count,
    ref System.byte PersistId,
    out System.int ErrorCode
)
```

C++/CLI

```
System.Object^ IGetObjectByPersistReference3(
    & System.int Count,
    & System.byte% PersistId,
    & [Out] System.int ErrorCode
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*Count*Size of persistent reference ID (see **Remarks**)*PersistId*Object's persistent reference ID (see **Remarks**)*ErrorCode*Success or error code as defined by [swPersistReferencedObjectStates_e](#) (see **Remarks**)

Return Value

Object (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.IGetObjectByPersistReference3](#).

Remarks

A persistent reference ID is related to a model. It is portable and can be saved within the model or in other places. Some persistent reference ID are general to all models and can be instantiated from all models. Your application must handle persistent reference IDs and their relationships among models.

Before calling this method:

1. Call [IModelDocExtension.GetPersistentReferenceCount3](#) to determine the size of the persistent reference ID.
2. Call [IModelDocExtension.IGetObjectByPersistReference3](#) to get the object's persistent reference ID.

The `swPersistReferencedObject_Suppressed` and `swPersistReferencedObject_Deleted` enumerators only apply to references of topological entities.

`IModelDocExtension::GetObjectByPersistReference3` was designed to return the base class of a feature to take advantage of the base feature class functionality. Objects returned by `IModelDocExtension::GetObjectByPersistReference3` are often features, including [IConfiguration](#), [IRefPlane](#), and [IRefAxis](#) objects. You must first obtain a reference to a feature, and then use [IFeature::GetSpecificFeature2](#) to get the original object.

See [Persistent Reference IDs](#) for more information.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetObjectByPersistReference3 Method](#)
[IModelDocExtension::GetPersistReference3 Method](#)
[IModelDocExtension::IsSamePersistentID Method](#)

Availability

SOLIDWORKS 2007 SP3, Revision Number 15.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetOLEObjects Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetOLEObjects Method (IModelDocExtension)

Get the OLE objects.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetOLEObjects( _  
    ByVal Options As System.Integer, _  
    ByVal OleObjectCount As System.Integer, _  
    ByRef LpOleObjects As SwOLEObject _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Options As System.Integer  
Dim OleObjectCount As System.Integer  
Dim LpOleObjects As SwOLEObject  
  
instance.IGetOLEObjects(Options, OleObjectCount, LpOleObjects)
```

C#

```
void IGetOLEObjects(  
    System.int Options,  
    System.int OleObjectCount,  
    out SwOLEObject LpOleObjects  
)
```

C++/CLI

```
void IGetOLEObjects(  
& System.int Options,  
& System.int OleObjectCount,  
& [Out] SwOLEObject^ LpOleObjects  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Options

Options as defined in [swOleObjectOptions](#) e

OleObjectCount

Number of OLE objects

LpOleObjects

- in-process, unmanaged C++: Pointer to an array of the [OLE objects](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IModelDocExtension::GetOLEObjectCount](#) to determine the size of the output array.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetOLEObjects Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetPersistReference Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetPersistReference Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::IGetPersistReference3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPersistReference( _  
    ByVal DipsObj As System.Object, _  
    ByVal Count As System.Integer _  
) As System.Byte
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim DipsObj As System.Object  
Dim Count As System.Integer  
Dim value As System.Byte  
  
value = instance.IGetPersistReference(DipsObj, Count)
```

C#

```
System.byte IGetPersistReference(  
    System.object DipsObj,  
    System.int Count  
)
```

C++/CLI

```
System.byte IGetPersistReference(  
& System.Object^ DipsObj,  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DipsObj

Count

Return Value

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetPersistReference](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetPersistReference3 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetPersistReference3 Method (IModelDocExtension)

Gets the persistent reference ID for the specified object in this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetPersistReference3( _  
    ByVal DipsObj As System.Object, _  
    ByVal Count As System.Integer _  
) As System.Byte
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim DipsObj As System.Object  
Dim Count As System.Integer  
Dim value As System.Byte  
  
value = instance.IGetPersistReference3(DipsObj, Count)
```

C#

```
System.byte IGetPersistReference3(  
    System.object DipsObj,  
    System.int Count  
)
```

C++/CLI

```
System.byte IGetPersistReference3(  
& System.Object^ DipsObj,  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DipsObj

Object

Count

Size of returned array assigned to that object (see [Remarks](#))

Return Value

Array containing the persistent reference ID assigned to that object

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetPersistReference3](#).

Example

[Get Object's Persistent Reference ID \(C++\)](#)

Remarks

Persistent reference ID values obtained using the now obsolete `IModelDocExtension::GetPersistReference` and its related obsolete methods, `IModelDocExtension::GetPersistReferenceCount` and `IModelDocExtension::GetObjectByPersistReference2`, are not compatible with persistent reference IDs obtained using this method and its related methods, [IModelDocExtension::GetPersistReferenceCount3](#) and [IModelDocExtension::GetObjectByPersistReference3](#) or [IModelDocExtension::IGetObjectByPersistReference3](#).

IMPORTANT: SOLIDWORKS recommends not using `IModelDocExtension::IGetPersistReference3` with `ModelDocExtension::GetPersistReferenceCount3` because when `ModelDocExtension::GetPersistReferenceCount3` is used with `ModelDocExtension::IGetPersistReference3`, you will experience a significant decrease in performance. Instead, use [IModelDocExtension::GetPersistReference3](#).

A persistent reference ID is related to a model. It is portable and can be saved within the model or in other places. Some persistent reference ID are general to all models and can be instantiated from all models. Your application must handle persistent reference IDs and their relationships among models.

The internal representations of the return value array may change, possibly from rebuild to rebuild, or more likely, from one release of SOLIDWORKS to the next, but their usage in finding the correct entity will be consistent across rebuilds and SOLIDWORKS releases.

To compare the referenced entities, you could use the Visual Basic **Is** operator to find out if the entities are the same.

See [Persistent Reference IDs](#) for more information.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::IsSamePersistentID Method](#)

Availability

SOLIDWORKS 2007 SP3, Revision Number 15.3

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetRenderMaterials Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetRenderMaterials Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::GetRenderMaterials2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRenderMaterials( _  
    ByVal Count As System.Integer _  
) As RenderMaterial
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Count As System.Integer  
Dim value As RenderMaterial
```

```
value = instance.IGetRenderMaterials(Count)
```

C#

```
RenderMaterial IGetRenderMaterials(  
    System.int Count  
)
```

C++/CLI

```
RenderMaterial^ IGetRenderMaterials(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of appearance

Return Value

- unmanaged C++: Pointer to an array of [appearances](#)
- VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [IModelDocExtension::GetRenderMaterialsCount](#) to get the value of Count.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IRenderMaterial Interface](#)
[IModelDocExtension::AddDefaultRenderMaterial Method](#)
[IModelDocExtension::AddRenderMaterial Method](#)
[IModelDocExtension::CreateRenderMaterial Method](#)
[IModelDocExtension::DeleteRenderMaterial Method](#)
[IModelDocExtension::GetMaterial Method](#)
[IModelDocExtension::GetRenderMaterials Method](#)
[IModelDocExtension::UpdateRenderMaterialsInSceneGraph Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSearchData Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetSearchData Method (IModelDocExtension)

Gets the SOLIDWORKS Search, third-party, application keywords from the model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub IGetSearchData( _  
    ByVal AppName As System.String, _  
    ByVal Count As System.Integer, _  
    ByRef AppNames As System.String, _  
    ByRef NodeNames As System.String, _  
    ByRef NodeValues As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim AppName As System.String  
Dim Count As System.Integer  
Dim AppNames As System.String  
Dim NodeNames As System.String  
Dim NodeValues As System.String  
  
instance.IGetSearchData(AppName, Count, AppNames, NodeNames, NodeValues)
```

C#

```
void IGetSearchData(  
    System.string AppName,  
    System.int Count,  
    out System.string AppNames,  
    out System.string NodeNames,  
    out System.string NodeValues  
)
```

C++/CLI

```

void IGetSearchData(
& System.String^ AppName,
& System.int Count,
& [Out] System.String^ AppNames,
& [Out] System.String^ NodeNames,
& [Out] System.String^ NodeValues
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppName

Third-party application name whose keywords to get

Count

Number of third-party application keywords

AppNames

- in-process, unmanaged C++: Pointer to an array of the third-party application names

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

NodeNames

- in-process, unmanaged C++: Pointer to an array of the third-party application name's keywords

- VBA, VB.NET, C#, and C++/CLI: Not supported

NodeValues

- in-process, unmanaged C++: Pointer to an array of the third-party application name's keyword values

- VBA, VB.NET, C#, and C++/CLI: Not supported

Return Value

Number of third-party application name's keywords

Remarks

Before calling this method, call [IModelDocExtension::GetSearchDataCount](#) to get the value of Count.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::AddOrUpdateSearchData Method](#)

[IModelDocExtension::DeleteSearchData Method](#)

[IModelDocExtension::GetSearchData Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetSectionProperties Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetSectionProperties Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::IGetSectionProperties2](#).

.NET Syntax

Visual Basic (Declaration)

```

Function IGetSectionProperties( _
    ByVal Count As System.Integer, _
    ByRef Sections As System.Object _
) As System.Double

```

Visual Basic (Usage)

```

Dim instance As IModelDocExtension
Dim Count As System.Integer
Dim Sections As System.Object
Dim value As System.Double

```

```
value = instance.IGetSectionProperties(Count, Sections)
```

C#




```
System.double IGetSectionProperties(  
    System.int Count,  
    ref System.object Sections  
)
```

C++/CLI

```
System.double IGetSectionProperties(  
& System.int Count,  
& System.Object^% Sections  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Sections

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetSectionProperties](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSectionProperties2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetSectionProperties2 Method (IModelDocExtension)

Gets the section properties for the following types of selected items:

- Planar model face in any document
- Face on a section plane
- Crosshatch section face in a section view in a drawing sketch
- Sketch

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSectionProperties2( _  
    ByVal Count As System.Integer, _  
    ByRef Sections As System.Object _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Count As System.Integer  
Dim Sections As System.Object  
Dim value As System.Double
```

```
value = instance.IGetSectionProperties2(Count, Sections)
```

C#

```
System.double IGetSectionProperties2(  
    System.int Count,  
    ref System.object Sections  
)
```

C++/CLI

```
System.double IGetSectionProperties2(
& System.int Count,
& System.Object^% Sections
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of sections
Sections
Array of sections

Return Value

- in-process, unmanaged C++: Pointer to an array of size 24 of the section properties for the selected items (see **Remarks**)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

This method clears the selection set.

If...	Then...
The user selected a set of either parallel planes or parallel faces	You can pass an empty sections array
The user selected any items and you pass a sections array	The properties of the user-selected items and the sections array are combined in the returned array
You pass a sections array and you do not want this array combined with the properties of any user-selected items	Clear any user-selected items

The objects in the Sections parameter are added to the current selection set. If the objects are already in the current selection set, an error is generated; that is, status code will be equal to 1, which means invalid input.

The format of the returned array (retval) of size 24:

• retval[0]	status of request: • 0 = success • 1 = invalid input • 2 = selected faces are not in the same or parallel planes • 3 = unable to compute section properties
• retval[1]	area
• retval[2]	centroid x
• retval[3]	centroid y
• retval[4]	centroid z
• retval[5]	moment of inertia XX
• reval[6]	moment of inertia YY
• retval[7]	moment of inertia ZZ
• retval[8]	moment of inertia -XY
• retval[9]	moment of inertia -ZX
• retval[10]	moment of inertia -YZ
• retval[11]	polar moment of inertia of an area at the centroid
• retval[12]	angle between principal axis and part axis
• retval[13]	principal moment of inertia of an area at the centroid, Ix

. retval[14]	principal moment of inertia of an area at the centroid, Iy
. retval[15]	direction vector X for principle axes (x component)
. retval[16]	direction vector X for principle axes (y component)
. retval[17]	direction vector X for principle axes (z component)
. retval[18]	direction vector Y for principle axes (x component)
. retval[19]	direction vector Y for principle axes (y component)
. retval[20]	direction vector Y for principle axes (z component)
. retval[21]	direction vector Z for principle axes (x component)
. retval[22]	direction vector Z for principle axes (y component)
. retval[23]	direction vector Z for principle axes (z component)

This method returns metric units unless explicitly specified otherwise.

 See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetSectionProperties2 Method](#)

 Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSheetMetalObjectsByPersistReference Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IGetSheetMetalObjectsByPersistReference Method (IModelDocExtension)

Gets the object, or objects, in a folded sheet metal part that correspond to the byte array of persistent reference IDs of an entity in a flattened sheet metal part.

 .NET Syntax

Visual Basic (Declaration)

```
Function IGetSheetMetalObjectsByPersistReference( _  
    ByVal Count As System.Integer, _  
    ByRef PersistId As System.Byte, _  
    ByRef ErrorCode As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Count As System.Integer  
Dim PersistId As System.Byte  
Dim ErrorCode As System.Integer  
Dim value As System.Object  
  
value = instance.IGetSheetMetalObjectsByPersistReference(Count, PersistId, ErrorCode)
```

C#

```
System.object IGetSheetMetalObjectsByPersistReference(  
    System.int Count,  
    ref System.byte PersistId,  
    out System.int ErrorCode  
)
```

C++/CLI

```
System.Object^ IGetSheetMetalObjectsByPersistReference(
& System.int Count,
& System.byte% PersistId,
[Out] System.int ErrorCode
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of persistent reference IDs

PersistId

Persistent reference IDs returned by [IModelDocExtension::GetFlattenSheetMetalPersistReference](#) or [IModelDocExtension::!GetFlattenSheetMetalPersistReference](#)

ErrorCode

Error as defined by [swPersistReferencedObjectStates_e](#)

Return Value

- in-process, unmanaged C++: Pointer to an array of objects in a folded sheet metal part that correspond to the byte array of persistent reference IDs of an entity in a flattened sheet metal part
- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IGetSheetMetalObjectsByPersistReference](#).

Remarks

See [Persistent Reference IDs](#) for more information.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::GetSheetMetalObjectsByPersistReference Method](#)

[IModelDocExtension::GetObjectByPersistReference3 Method](#)

[IModelDocExtension::GetPersistReference3 Method](#)

[IModelDocExtension::GetPersistReferenceCount3 Method](#)

[IModelDocExtension::!GetObjectByPersistReference3 Method](#)

[IModelDocExtension::!GetPersistReference3 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IListExternalFileReferences Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IListExternalFileReferences Method (IModelDocExtension)

Gets the names and statuses of the external file references on this part or assembly.

.NET Syntax

Visual Basic (Declaration)

```
Sub IListExternalFileReferences( _
    ByVal NumRefs As System.Integer, _
    ByRef ModelPathName As System.String, _
    ByRef CompPathName As System.String, _
    ByRef Feature As System.String, _
    ByRef DataType As System.String, _
    ByRef Status As System.Integer, _
    ByRef RefEntity As System.String, _
    ByRef FeatComp As System.String, _
    ByRef ConfigOption As System.Integer, _
    ByRef ConfigName As System.String _
)
```

```

Dim instance As IModelDocExtension
Dim NumRefs As System.Integer
Dim ModelPathName As System.String
Dim CompPathName As System.String
Dim Feature As System.String
Dim DataType As System.String
Dim Status As System.Integer
Dim RefEntity As System.String
Dim FeatComp As System.String
Dim ConfigOption As System.Integer
Dim ConfigName As System.String

```

instance.IListExternalFileReferences(NumRefs, ModelPathName, CompPathName, Feature, DataType, Status, RefEntity, FeatComp, ConfigOption, ConfigName)

C#

```

void IListExternalFileReferences(
    System.int NumRefs,
    out System.string ModelPathName,
    out System.string CompPathName,
    out System.string Feature,
    out System.string DataType,
    out System.int Status,
    out System.string RefEntity,
    out System.string FeatComp,
    out System.int ConfigOption,
    out System.string ConfigName
)

```

C++/CLI

```

void IListExternalFileReferences(
    & System.int NumRefs,
    & [Out] System.String^ ModelPathName,
    & [Out] System.String^ CompPathName,
    & [Out] System.String^ Feature,
    & [Out] System.String^ DataType,
    & [Out] System.int Status,
    & [Out] System.String^ RefEntity,
    & [Out] System.String^ FeatComp,
    & [Out] System.int ConfigOption,
    & [Out] System.String^ ConfigName
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumRefs

Number of external references

ModelPathName

- in-process, unmanaged C++: Pointer to an array of path names of documents of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

CompPathName

- in-process, unmanaged C++: Pointer to an array of path names of referenced components of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

Feature

- in-process, unmanaged C++: Pointer to an array of in-context items (sketches, features, and so on) of size NumRefs

• VBA, VB.NET, C#, and C++/CLI: Not supported

DataType

- in-process, unmanaged C++: Pointer to an array of the type data used to create the items (converted edge or face, converted or offset sketch entity, body, and so on) of size NumRef

• VBA, VB.NET, C#, and C++/CLI: Not supported

Status

- in-process, unmanaged C++: Array of statuses of the external references as defined in [swExternalReferenceStatus_e](#)

• VBA, VB.NET, C#, and C++/CLI: Not supported

RefEntity

- in-process, unmanaged C++: Pointer to an array of actual items being used and the names of the documents that contain the items of size NumRefs
- VBA, VB.NET, C#, and C++/CLI: Not supported

FeatComp

- in-process, unmanaged C++: Pointer to an array of the names of the components in which the affected features exist of size NumRefs; this information is only displayed when one or more RefEntity is in a different component in an assembly and does not apply to derived parts
- VBA, VB.NET, C#, and C++/CLI: Not supported

ConfigOption

Configuration option as defined by [swExternalFileReferencesConfig_e](#)

ConfigName

Name of the configuration when configOption is swExternalFileReferencesNamedConfig

Remarks

Before calling this method, call [IModelDocExtension::ListExternalFileReferencesCount](#) to specify NumRefs.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)
[IModelDoc2::BreakAllExternalReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)
[IModelDoc2::LockAllExternalReferences Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMultiSelect Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IMultiSelect Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::MultiSelect2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IMultiSelect( _  
    ByVal Count As System.Integer, _  
    ByRef Objects As System.Object, _  
    ByVal AppendFlag As System.Boolean, _  
    ByVal Data As SelectData _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Count As System.Integer  
Dim Objects As System.Object  
Dim AppendFlag As System.Boolean  
Dim Data As SelectData  
Dim value As System.Integer  
  
value = instance.IMultiSelect(Count, Objects, AppendFlag, Data)
```

C#

```
System.int IMultiSelect(  
    System.int Count,  
    ref System.object Objects,  
    System.bool AppendFlag,  
    SelectData Data  
)
```

C++/CLI

```
System.int IMultiSelect(  
& System.int Count,  
& System.Object^% Objects,  
& System.bool AppendFlag,  
& SelectData^ Data  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of selectable objects

Objects

Array of selectable objects:

- objects can be heterogeneous
- if any object is not selected, then it is ignored

AppendFlag

True to append the objects to the selection list, false replace the current selection list with these objects

Data

[ISelectData](#) object

Return Value

Number of objects selected

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IMultiSelect](#).

Remarks

If this now obsolete method is used to populate a selection box in a PropertyManager page, then [IPropertyManagerPage2Handler9::OnSubmitSelection](#) only fires when set to true and [ISelectData::Mark](#) is set to 0.

Use [IModelDocExtension::MultiSelect2](#) if you need [IPropertyManagerPage2Handler9::OnSubmitSelection](#) to fire when set to true and [ISelectData::Mark](#) is set to either a 0 or non-0 value.

The [Callout](#) property of the [ISelectData](#) object is not used by this method.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::MultiSelect Method](#)
[IModelDocExtension::SelectById2 Method](#)
[ISelectionMgr Interface](#)
[IModelDocExtension::SelectAll Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IncludeMassPropertiesOfHiddenBodies Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : IncludeMassPropertiesOfHiddenBodies Property (IModelDocExtension)

Gets or sets whether to include the mass properties of hidden components in the assembly.

.NET Syntax

Visual Basic (Declaration)

```
Property IncludeMassPropertiesOfHiddenBodies As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean
```

```
instance.IncludeMassPropertiesOfHiddenBodies = value
```

```
value = instance.IncludeMassPropertiesOfHiddenBodies
```

C#

```
System.bool IncludeMassPropertiesOfHiddenBodies {get; set;}
```

C++/CLI

```
property System.bool IncludeMassPropertiesOfHiddenBodies {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to include the mass properties of hidden components in the assembly, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::IncludeMassPropertiesOfHiddenBodies](#).

Example

[Get Mass Properties of Visible and Hidden Components \(C#\)](#)
[Get Mass Properties of Visible and Hidden Components \(VB.NET\)](#)
[Get Mass Properties of Visible and Hidden Components \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IAssemblyDoc::HasUnloadedComponents Method](#)
[IAssemblyDoc::HideComponent Method](#)
[IAssemblyDoc::IsSetComponentVisibility Method](#)
[IAssemblyDoc::SetComponentVisibility Method](#)
[IComponent2::HasUnloadedComponents Method](#)
[IComponent2::IsHidden Method](#)
[IModelDoc2::HideComponent2 Method](#)
[IModelDocExtension::CreateMassProperty Method](#)

Availability

SOLIDWORKS 2009 SP03, Revision Number 17.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertAnnotationFavorite Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertAnnotationFavorite Method (IModelDocExtension)

Inserts annotations from the specified favorite file at the specified location.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertAnnotationFavorite( _
    ByVal BstrFileName As System.String, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As Annotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim BstrFileName As System.String
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As Annotation
```

```
value = instance.InsertAnnotationFavorite(BstrFileName, X, Y, Z)
```

C#

```
Annotation InsertAnnotationFavorite(
    System.string BstrFileName,
```



```

        System.double X,
        System.double Y,
        System.double Z
    )

```

C++/CLI

```

Annotation^ InsertAnnotationFavorite(
&    System.String^ BstrFileName,
&    System.double X,
&    System.double Y,
&    System.double Z
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BstrFileName

Path and filename of favorites file

X

x coordinate where to insert the annotations

Y

y coordinate where to insert the annotations

Z

z coordinate where to insert the annotations

Return Value

[Annotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertAnnotationFavorite](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertAnnotationView Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertAnnotationView Method (IModelDocExtension)

Inserts an annotation view in this part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```

Function InsertAnnotationView( _
    ByVal AnnotationViewingDirection As System.Integer, _
    ByVal DirectionReference As System.Object, _
    ByVal FlipDirection As System.Boolean, _
    ByVal HorizontalDirectionReference As System.Object, _
    ByVal AngleMadeWithHorizontal As System.Integer _
) As AnnotationView

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim AnnotationViewingDirection As System.Integer
Dim DirectionReference As System.Object
Dim FlipDirection As System.Boolean
Dim HorizontalDirectionReference As System.Object

```

```
Dim AngleMadeWithHorizontal As System.Integer
Dim value As AnnotationView

value = instance.InsertAnnotationView(AnnotationViewingDirection, DirectionReference,
FlipDirection, HorizontalDirectionReference, AngleMadeWithHorizontal)
```

C#

```
AnnotationView InsertAnnotationView(
    System.int AnnotationViewingDirection,
    System.object DirectionReference,
    System.bool FlipDirection,
    System.object HorizontalDirectionReference,
    System.int AngleMadeWithHorizontal
)
```

C++/CLI

```
AnnotationView^ InsertAnnotationView(
    & System.int AnnotationViewingDirection,
    & System.Object^ DirectionReference,
    & System.bool FlipDirection,
    & System.Object^ HorizontalDirectionReference,
    & System.int AngleMadeWithHorizontal
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AnnotationViewingDirection

Defined by either any [swStandardViews_e](#) enumerator or 0 for selection

DirectionReference

If 0 specified for AnnotationViewingDirection, then specify a [face](#) or [plane](#) to define the direction of the annotation view

FlipDirection

True to flip the annotation view in the opposite direction, false to not

HorizontalDirectionReference

An [edge](#), [sketch](#), or [face](#)

AngleMadeWithHorizontal

Angle (in degrees) with the specified HorizontalDirectionReference

Return Value

Newly inserted [annotation view](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertAnnotationView](#).

Example

'VBA

'-----

' Preconditions:

- ' 1. Open a SOLIDWORKS model file.
- ' 2. Select a face to define the direction of the annotation view.
- ' 3. Select an edge as a horizontal direction reference.

'

' Postconditions: Observe that a new annotation view is inserted at a 45 degree angle in reference to the face and edge.

'-----

```
Dim swApp As SldWorks.SldWorks
```

```
Dim swModel As SldWorks.ModelDoc2
```

```
Dim swModExt As SldWorks.ModelDocExtension
```

```
Dim swSelMgr As SldWorks.SelectionMgr
```

```
Dim swFace As SldWorks.Face2
```

```
Dim swEdge As SldWorks.edge
```

```
Dim swAnnoView As SldWorks.AnnotationView
```

```

Dim i As Integer
Sub main()
    Set swApp = Application.SldWorks
    Set swModel = swApp.ActiveDoc
    Set swModExt = swModel.Extension
    Set swSelMgr = swModel.SelectionManager
    For i = 0 To swSelMgr.GetSelectedObjectCount2(-1) - 1
        If swSelMgr.GetSelectedObjectType3(i + 1, -1) = swSelFACES Then
            Set swFace = swSelMgr.GetSelectedObject6(i + 1, -1)
        End If
        If swSelMgr.GetSelectedObjectType3(i + 1, -1) = swSelEDGES Then
            Set swEdge = swSelMgr.GetSelectedObject6(i + 1, -1)
        End If
    Next i

    Set annoView = swModExt.InsertAnnotationView(0, Nothing, False, swEdge, 45)
    swModel.EditRebuild3
End Sub

```

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertAttachment Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertAttachment Method (IModelDocExtension)

Inserts a file as an Attachment to this document.

.NET Syntax

Visual Basic (Declaration)

```

Function InsertAttachment( _
    ByVal FileName As System.String, _
    ByVal Linked As System.Boolean _
) As System.Boolean

```

Visual Basic (Usage)

```

Dim instance As IModelDocExtension
Dim FileName As System.String
Dim Linked As System.Boolean
Dim value As System.Boolean

value = instance.InsertAttachment(FileName, Linked)

```

C#



```
System.bool InsertAttachment(  
    System.string FileName,  
    System.bool Linked  
)
```

C++/CLI

```
System.bool InsertAttachment(  
& System.String^ FileName,  
& System.bool Linked  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FileName

Path and filename of file to insert as an Attachment to this document

Linked

True to link the document to the file, false to not

Return Value

True if the file is inserted as an Attachment, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertAttachment](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::DeleteAttachment Method](#)

[IModelDocExtension::GetAttachmentCount Method](#)

[IModelDocExtension::GetAttachments Method](#)

[IModelDocExtension::IGetAttachments Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBOMBalloon Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertBOMBalloon Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::InsertBomBalloon2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBOMBalloon( _  
    ByVal Style As System.Integer, _  
    ByVal Size As System.Integer, _  
    ByVal UpperTextStyle As System.Integer, _  
    ByVal UpperText As System.String, _  
    ByVal LowerTextStyle As System.Integer, _  
    ByVal LowerText As System.String, _  
    ByVal CustomSize As System.Double, _  
    ByVal ShowQuantity As System.Boolean, _  
    ByVal QuantityPlacement As System.Short, _  
    ByVal QuantityDenotationText As System.String _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Style As System.Integer
```

```

Dim Size As System.Integer
Dim UpperTextStyle As System.Integer
Dim UpperText As System.String
Dim LowerTextStyle As System.Integer
Dim LowerText As System.String
Dim CustomSize As System.Double
Dim ShowQuantity As System.Boolean
Dim QuantityPlacement As System.Short
Dim QuantityDenotationText As System.String
Dim value As System.Object

```

```

value = instance.InsertBOMBalloon(Style, Size, UpperTextStyle, UpperText, LowerTextStyle,
LowerText, CustomSize, ShowQuantity, QuantityPlacement, QuantityDenotationText)

```

C#

```

System.Object InsertBOMBalloon(
    System.Int Style,
    System.Int Size,
    System.Int UpperTextStyle,
    System.String UpperText,
    System.Int LowerTextStyle,
    System.String LowerText,
    System.Double CustomSize,
    System.Boolean ShowQuantity,
    System.Short QuantityPlacement,
    System.String QuantityDenotationText
)

```

C++/CLI

```

System.Object^ InsertBOMBalloon(
    & System.Int Style,
    & System.Int Size,
    & System.Int UpperTextStyle,
    & System.String^ UpperText,
    & System.Int LowerTextStyle,
    & System.String^ LowerText,
    & System.Double CustomSize,
    & System.Boolean ShowQuantity,
    & System.Short QuantityPlacement,
    & System.String^ QuantityDenotationText
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Style

Balloon style as defined in [swBalloonStyle](#) e

Size

Balloon size as defined in [swBalloonFit](#) e

UpperTextStyle

Style for the upper text of the balloon as defined in [swBalloonTextContent](#) e (see **Remarks**)

UpperText

Upper text of the balloon

LowerTextStyle

Style for the lower text of the balloon as defined in [swBalloonTextContent](#) e; valid for balloons only and Style must be set to swBS_SplitCirc (see **Remarks**)

LowerText

Lower text of the balloon; valid for balloons only and Style must be set to swBS_SplitCirc

CustomSize

User-defined size of the balloon; Size must be set to swBF_UserDef

ShowQuantity

True to show quantity, false to not

QuantityPlacement

Placement of quantity value:

- 0 = Left
- 1 = Right
- 2 = Top
- 3 = Bottom

QuantityDenotationText

Denotation text for quantity

Return Value

[Note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertBOMBalloon](#).

Remarks

See [INote::PropertyLinkedText](#) for examples of link strings usable with swBalloonTextContent_e.swBalloonTextCustomProperties.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IDrawingDoc::AutoBalloon3 Method](#)
[INote::GetBomBalloonText Method](#)
[INote::GetBomBalloonTextStyle Method](#)
[INote::SetBomBalloonText Method](#)
[INote::IsBomBalloon Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertBOMBalloon2 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertBOMBalloon2 Method (IModelDocExtension)

Inserts a BOM balloon for the selected item.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function InsertBOMBalloon2( _  
    ByVal BalloonOptions As BalloonOptions _  
) As Note
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim BalloonOptions As BalloonOptions  
Dim value As Note  
  
value = instance.InsertBOMBalloon2(BalloonOptions)
```

C#

```
Note InsertBOMBalloon2(  
    BalloonOptions BalloonOptions  
)
```

C++/CLI

```
Note^ InsertBOMBalloon2(  
& BalloonOptions^ BalloonOptions  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BalloonOptions
[IBalloonOptions](#)

Return Value

[Note](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertBOMBalloon2](#).

Example

[Insert and Show BOM Table and BOM Balloon \(VBA\)](#)
[Insert and Show BOM Table and BOM Balloon \(VB.NET\)](#)
[Insert and Show BOM Table and BOM Balloon \(C#\)](#)

Remarks

To create a single BOM balloon:

1. Select an item for which to create a BOM balloon.
2. Call [IModelDocExtension::CreateBalloonOptions](#) to create an [IBalloonOptions](#) object.
3. Set the properties on the IBalloonOptions object.
4. Call this method using the IBalloonOptions object in the BalloonOptions argument.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::InsertStackedBalloon2 Method](#)
[IDrawingDoc::AutoBalloon5 Method](#)
[INote::GetBalloonOptions Method \(\)](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBomTable Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertBomTable Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::InsertBomTable2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBomTable( _  
    ByVal TemplateName As System.String, _  
    ByVal X As System.Integer, _  
    ByVal Y As System.Integer, _  
    ByVal BomType As System.Integer, _  
    ByVal ConfigurationName As System.String _  
    ) As BomTableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim TemplateName As System.String  
Dim X As System.Integer  
Dim Y As System.Integer  
Dim BomType As System.Integer  
Dim ConfigurationName As System.String  
Dim value As BomTableAnnotation  
  
value = instance.InsertBomTable(TemplateName, X, Y, BomType, ConfigurationName)
```

C#

```
BomTableAnnotation InsertBomTable(  
    System.string TemplateName,  
    System.int X,  
    System.int Y,  
    System.int BomType,  
    System.string ConfigurationName  
)
```

```
BomTableAnnotation^ InsertBomTable(
& System.String^ TemplateName,
& System.int X,
& System.int Y,
& System.int BomType,
& System.String^ ConfigurationName
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

Path and name of BOM table template (see **Remarks**)

X

X coordinate for the placement of the BOM table

Y

Y coordinate for the placement of the BOM table

BomType

Type of BOM table as defined by [swBomType_e](#)

ConfigurationName

Name of the configuration for this BOM table (see **Remarks**)

Return Value

[BOM table annotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertBomTable](#).

Example

[Insert BOM Table and Stacked Balloon \(VB.NET\)](#)

[Insert BOM Table and Stacked Balloon \(VBA\)](#)

Remarks

The system will not default to the default configuration when you specify an empty string for Configuration. You must specify the configuration.

By default, the BOM table templates are in the <install_dir>\lang\<language> folder and have a filename extension of **.sldbomtbt**. The template and table must be of the same type. For example, you could specify **C:\Program Files\SOLIDWORKS\lang\English\bom-standard.sldbomtbt** for *TemplateName* if you wanted to insert an English-version of the standard BOM table template.

If the BOM table is a parts-only or indented-style BOM and *ConfigurationName* is invalid, then the BOM is not created.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IView::InsertBomTable2 Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBomTable2 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : [InsertBomTable2 Method \(IModelDocExtension\)](#)

Obsolete. Superseded by [IModelDocExtension::InsertBomTable3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBomTable2( _
    ByVal TemplateName As System.String, _
    ByVal X As System.Integer, _
    ByVal Y As System.Integer, _
```



```

    ByVal BomType As System.Integer, _
    ByVal ConfigurationName As System.String, _
    ByVal Hidden As System.Boolean _
) As BomTableAnnotation

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim TemplateName As System.String
Dim X As System.Integer
Dim Y As System.Integer
Dim BomType As System.Integer
Dim ConfigurationName As System.String
Dim Hidden As System.Boolean
Dim value As BomTableAnnotation

value = instance.InsertBomTable2(TemplateName, X, Y, BomType, ConfigurationName, Hidden)

```

C#

```

BomTableAnnotation InsertBomTable2(
    System.string TemplateName,
    System.int X,
    System.int Y,
    System.int BomType,
    System.string ConfigurationName,
    System.bool Hidden
)

```

C++/CLI

```

BomTableAnnotation^ InsertBomTable2(
    & System.String^ TemplateName,
    & System.int X,
    & System.int Y,
    & System.int BomType,
    & System.String^ ConfigurationName,
    & System.bool Hidden
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName
Path and name of BOM table template (see **Remarks**)

X
X coordinate for the placement of the BOM table

Y
Y coordinate for the placement of the BOM table

BomType
Type of BOM table as defined by [swBomType_e](#)

ConfigurationName
Name of the configuration for this BOM table (see **Remarks**)

Hidden
True to hide the BOM table, false to show it

Return Value

[BOM table annotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertBomTable2](#).

Remarks

The system does not default to the Default configuration when you specify an empty string for Configuration. You must specify the configuration.

BOM table templates are in the <SOLIDWORKS_install_dir>\lang\<language> folder and have a filename extension of .sldbomtbt. The template and table must be of the same type. For example, you could specify C:\Program Files\SOLIDWORKS\lang\English\bom-standard.sldbomtbt for TemplateName if you wanted to insert an English-version of the standard BOM table template.

If the BOM table is a parts-only or indented-style BOM and ConfigurationName is invalid, then the BOM is not created.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IView::InsertBomTable3 Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

InsertBomTable3 Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertBomTable3 Method (IModelDocExtension)

Obsolete. Superseded by [IModelDocExtension::InsertBomTable4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBomTable3( _  
    ByVal TemplateName As System.String, _  
    ByVal X As System.Integer, _  
    ByVal Y As System.Integer, _  
    ByVal BomType As System.Integer, _  
    ByVal ConfigurationName As System.String, _  
    ByVal Hidden As System.Boolean, _  
    ByVal IndentedNumberingType As System.Integer, _  
    ByVal DetailedCutList As System.Boolean _  
) As BomTableAnnotation
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim TemplateName As System.String  
Dim X As System.Integer  
Dim Y As System.Integer  
Dim BomType As System.Integer  
Dim ConfigurationName As System.String  
Dim Hidden As System.Boolean  
Dim IndentedNumberingType As System.Integer  
Dim DetailedCutList As System.Boolean  
Dim value As BomTableAnnotation  
  
value = instance.InsertBomTable3(TemplateName, X, Y, BomType, ConfigurationName, Hidden,  
    IndentedNumberingType, DetailedCutList)
```

C#

```
BomTableAnnotation InsertBomTable3(  
    System.string TemplateName,  
    System.int X,  
    System.int Y,  
    System.int BomType,  
    System.string ConfigurationName,  
    System.bool Hidden,  
    System.int IndentedNumberingType,  
    System.bool DetailedCutList  
)
```

C++/CLI

```
BomTableAnnotation^ InsertBomTable3(  
& System.String^ TemplateName,  
& System.int X,  
& System.int Y,  
& System.int BomType,
```

```
& System.String^ ConfigurationName,  
& System.bool Hidden,  
& System.int IndentedNumberingType,  
& System.bool DetailedCutList  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

Path and name of BOM table template (see **Remarks**)

X

X coordinate for the placement of the BOM table

Y

Y coordinate for the placement of the BOM table

BomType

Type of BOM table as defined by [swBomType_e](#)

ConfigurationName

Name of the configuration for this BOM table (see **Remarks**)

Hidden

True to hide the BOM table, false to show it

IndentedNumberingType

Type of numbering as defined by [swNumberingType_e](#); valid only if BomType = swBomType_e.swBomType_Indented

DetailedCutList

True to show the detailed cut list, false to not; valid only if BomType = swBomType_e.swBomType_Indented

Return Value

[BOM table annotation](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertBomTable3](#).

Remarks

The system does not default to the Default configuration when you specify an empty string for Configuration. You must specify the configuration.

BOM table templates are in the <SOLIDWORKS_install_dir>\lang\<language> folder and have a filename extension of .sldbomtbt. The template and table must be of the same type. For example, you could specify C:\Program Files\SOLIDWORKS\lang\English\bom-standard.sldbomtbt for TemplateName if you wanted to insert an English-version of the standard BOM table template.

If the BOM table is a parts-only or indented-style BOM and ConfigurationName is invalid, then the BOM is not created.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IView::InsertBomTable4 Method](#)

[IModelDocExtension::InsertGeneralTableAnnotation Method\(\)](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertBomTable4 Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertBomTable4 Method (IModelDocExtension)

Inserts a bill of materials (BOM) table in a part or assembly document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertBomTable4( _  
    ByVal TemplateName As System.String, _  
    ByVal X As System.Integer, _  
    ByVal Y As System.Integer, _  
    ByVal BomType As System.Integer, _  
    ByVal ConfigurationName As System.String, _  
    ByVal Hidden As System.Boolean, _  
    ByVal IndentedNumberingType As System.Integer, _  
    ByVal DetailedCutList As System.Boolean, _
```

```
ByVal DissolvePartLevelRows As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim TemplateName As System.String  
Dim X As System.Integer  
Dim Y As System.Integer  
Dim BomType As System.Integer  
Dim ConfigurationName As System.String  
Dim Hidden As System.Boolean  
Dim IndentedNumberingType As System.Integer  
Dim DetailedCutList As System.Boolean  
Dim DissolvePartLevelRows As System.Boolean  
Dim value As System.Object  
  
value = instance.InsertBomTable4(TemplateName, X, Y, BomType, ConfigurationName, Hidden,  
IndentedNumberingType, DetailedCutList, DissolvePartLevelRows)
```

C#

```
System.object InsertBomTable4(  
    System.string TemplateName,  
    System.int X,  
    System.int Y,  
    System.int BomType,  
    System.string ConfigurationName,  
    System.bool Hidden,  
    System.int IndentedNumberingType,  
    System.bool DetailedCutList,  
    System.bool DissolvePartLevelRows  
)
```

C++/CLI

```
System.Object^ InsertBomTable4(  
& System.String^ TemplateName,  
& System.int X,  
& System.int Y,  
& System.int BomType,  
& System.String^ ConfigurationName,  
& System.bool Hidden,  
& System.int IndentedNumberingType,  
& System.bool DetailedCutList,  
& System.bool DissolvePartLevelRows  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TemplateName

Path and name of BOM table template (see **Remarks**)

X

X coordinate for the placement of the BOM table

Y

Y coordinate for the placement of the BOM table

BomType

Type of BOM table as defined by [swBomType_e](#)

ConfigurationName

Name of the configuration for this BOM table (see **Remarks**)

Hidden

True to hide the BOM table, false to show it

IndentedNumberingType

Type of numbering as defined by [swNumberingType_e](#); valid only if BomType = swBomType_e.swBomType_Indented

DetailedCutList

True to show the detailed cut list, false to not

DissolvePartLevelRows

True to dissolve part level rows, false to not; valid only if DetailedCutList is true

Return Value

[BOM table annotation](#)

Example

[Insert and Show BOM Table in Assembly \(VBA\)](#)
[Insert and Show BOM Table in Assembly \(VB.NET\)](#)
[Insert and Show BOM Table in Assembly \(C#\)](#)

Remarks

The system does not default to the Default configuration when you specify an empty string for Configuration. You must specify the configuration.

BOM table templates are in the <SOLIDWORKS_install_dir>\lang\<language> folder and have a filename extension of .sldbombt. The template and table must be of the same type. For example, you could specify **C:\Program Files\SOLIDWORKS\lang\English\bom-standard.sldbombt** for TemplateName if you wanted to insert an English-version of the standard BOM table template.

If the BOM table is a parts-only or indented-style BOM and ConfigurationName is invalid, then the BOM is not created.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IView::InsertBomTable5 Method \(\)](#)
[IModelDocExtension::InsertGeneralTableAnnotation Method \(\)](#)
[IBomFeature::DissolvePartLevelRows Property \(\)](#)

Availability

SOLIDWORKS 2024 FCS, Revision Number 32

SOLIDWORKS API Help

[Send comments on this topic.](#)

InsertCamera Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : InsertCamera Method (IModelDocExtension)

Inserts a camera in this document.

.NET Syntax

Visual Basic (Declaration)

```
Function InsertCamera() As Camera
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As Camera
```

```
value = instance.InsertCamera()
```

C#

```
Camera InsertCamera()
```

C++/CLI

```
Camera^ InsertCamera();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Camera](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::InsertCamera](#).

Example

[Insert Camera \(C#\)](#)
[Insert Camera \(VB.NET\)](#)
[Insert Camera \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0